

Algorithms, AI & Data Science II

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Notes:

- **Lecture L11: Bayesian Inference and Learning** 16.06.2025
- **Educational objective:** This week's lecture introduces foundations of Bayesian learning. We introduce the Bayesian view on statistical inference and reconsider linear regression from a statistical inference perspective.
 - Bayesian inference
 - Statistical inference perspective on linear regression
- **Exercise sheet 06:** Hypothesis Testing and Causal Diagrams due 23.06.2025
- **Handout version:** 2025/06/03 15:03:40

Motivation

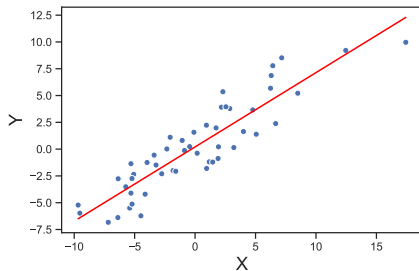
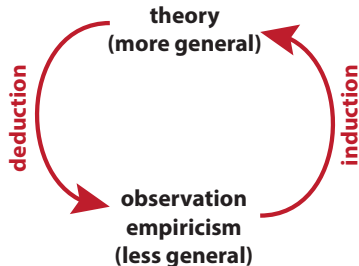
- ▶ statistical modelling and inference helps us to **draw conclusions** based on (uncertain) data
- ▶ example: we observe $x = 57$ heads in $n = 100$ coin tosses, model parameter p ?

$$\mathcal{L}(p|x) = \binom{n}{x} p^x (1-p)^{n-x}$$

- ▶ **maximum likelihood** estimation yields → L10

$$\hat{p} = \frac{x}{n} = 0.57$$

- ▶ so far, we have taken a **frequentist's view** on statistical inference



Notes:

Limits of frequentist inference

- ▶ with **frequentist interpretation of probability**, $P(X = x)$ is normalised frequency of outcome x in **repeated experiments**
- ▶ difficult if we only have **few observations** or must argue about **rare events**

thought experiment

→ Laplace, 1814

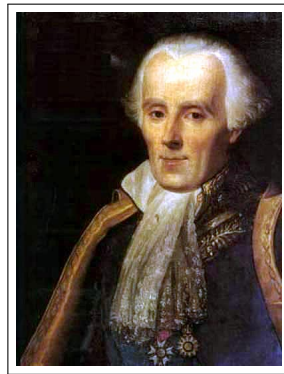
What is the **probability that the sun rises** tomorrow ($X = 1$)?

1. prior to first sunrise: $P(X = 1) = 0$ (sun surely not rising)
2. thereafter: $P(X = 1) = 1$ thereafter (sun surely rising)

Laplace's rule of succession

Assuming we have observed n sunrises so far we estimate $P(X = 1) = \frac{n+1}{n+2}$

- ▶ we use observations to progressively **update our belief** about future events



Pierre-Simon LaPlace

1749 – 1827

Notes:

Bayesian inference

- ▶ **Bayesian interpretation** of probabilities
 - ▶ $P(X = x)$ is our **belief** that X assumes value x
 - ▶ quantifies our **uncertainty** about a phenomenon
- ▶ **Bayesian learning**
 - ▶ we treat model parameter Θ as random variable
 - ▶ prior distribution $P(\Theta)$ models our **uncertainty** about Θ (prior to observing any data)
 - ▶ we use observations x to **update prior** $P(\Theta)$ to posterior distribution $P(\Theta|x)$

Bayesian update rule

Bayes theorem relates **posterior probability** $P(\Theta|x)$ to **prior probability** $P(\Theta)$ and **likelihood of observation** $P(x|\Theta)$

$$P(\Theta|x) = \frac{P(x|\Theta)}{P(x)} \cdot P(\Theta)$$

where $P(x) = \int_{\Theta} P(x|\Theta)P(\Theta)$ is normalising **marginal likelihood**



Thomas Bayes
1702 – 1761

Notes:

Beta distribution

- ▶ to model parameters of statistical models, we consider a **versatile family of continuous probability distributions** on $[0, 1]$

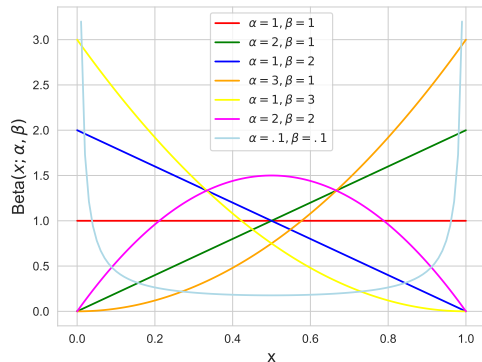
Beta distribution

- ▶ continuous distribution on interval $[0, 1]$

$$\text{Beta}(x; \alpha, \beta) = \frac{x^{\alpha-1}(1-x)^{\beta-1}}{B(\alpha, \beta)}$$

with Euler's beta function $B(\alpha, \beta) := \int_0^1 x^{\alpha-1}(1-x)^{\beta-1} dx$

- ▶ mean: $\frac{\alpha}{\alpha+\beta}$
- ▶ variance: $\frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}$
- ▶ $\alpha = \beta = 1$: uniform distribution on $[0, 1]$
- ▶ $\alpha = \beta$: symmetric distributions
- ▶ $\alpha < \beta$ positive skewness, $\alpha > \beta$ negative skewness
- ▶ mode (max. value): $\frac{\alpha-1}{\alpha+\beta-2}$



Beta distribution for different parameters α and β

Notes:

Group Exercise 11-01

1/2

- Consider the coin tossing example from L10, where you observe an increasing number of n coin tosses and where p is unknown. Starting from a uniform prior distribution for p , use Bayesian inference to calculate the a posterior probability of p after observing x heads in n tosses.

- We assume that we observe x heads in n coin tosses (i.e. x is our data). We want to estimate the **unknown** model parameter p , which is the probability that a **coin toss** shows heads.
- prior to observing any outcome, we treat p as **uniformly distributed parameter**, i.e. as prior distribution we use $P(p) = \text{Beta}(p; 1, 1)$
- we use Bayes rule to **update our belief** after observing x times heads in n coin tosses, where $P(x|p)$ is given by the Binomial distribution

$$P(p|x) = \frac{P(x|p)P(p)}{P(x)} = \frac{\binom{n}{x} p^x (1-p)^{n-x} \text{Beta}(p; 1, 1)}{P(x)}$$

- we calculate the **marginal distribution** $P(x)$ as follows

$$P(x) = \int_0^1 P(x|p)P(p)dp = \int_0^1 \binom{n}{x} p^x (1-p)^{n-x} dp$$

Notes:

- ▶ The **posterior distribution** of parameter p after observing x heads in n coin tosses is

$$P(p|k) = \frac{\binom{n}{x} p^x (1-p)^{n-x} \text{Beta}(p; 1, 1)}{\binom{n}{x} \int_0^1 p^x (1-p)^{n-x} dp}$$

- ▶ Using the **definition of Euler's Beta function** $B(\alpha, \beta)$ we have

$$P(p|k) = \frac{p^x (1-p)^{n-x} \text{Beta}(p; 1, 1)}{B(x+1, n-x+1)}$$

- ▶ we can apply the **definition of the Beta distribution** $\text{Beta}(\alpha, \beta)$ and obtain

$$P(p|k) = \text{Beta}(p; 1, 1) \text{Beta}(p; x+1, n-x+1)$$

- ▶ with $\text{Beta}(p; 1, 1) = 1$ for any $p \in [0, 1]$ we obtain the **posterior distribution of p**

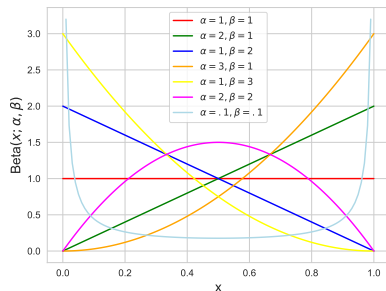
$$P(p|k) = \text{Beta}(p; x+1, n-x+1)$$

- ▶ observation of x heads in n coin tosses leads to **update of the parameters of the Beta prior**

Notes:

Beta distribution as conjugate prior

- ▶ in our example we started with uniform prior $P(\Theta) \sim \text{Beta}(\Theta; 1, 1)$, i.e. prior distribution is from **family of Beta distributions**
- ▶ updating prior based on likelihood $P(x|\Theta)$, we arrived at $P(\Theta|x) \sim \text{Beta}(\Theta; x + 1, n - x + 1)$
- ▶ i.e. posterior distribution is again from family of Beta distributions
- ▶ if prior $P(\Theta)$ and posterior $P(\Theta|x)$ are from same family we call $P(\Theta)$ **conjugate prior for $P(x|\Theta)$**
- ▶ Beta distribution is conjugate prior for **Bernoulli, Binomial, negative Binomial, and geometric distribution**
- ▶ use of conjugate priors makes it convenient to obtain **analytical expression** for posterior → exercise sheet



Bayesian update rule

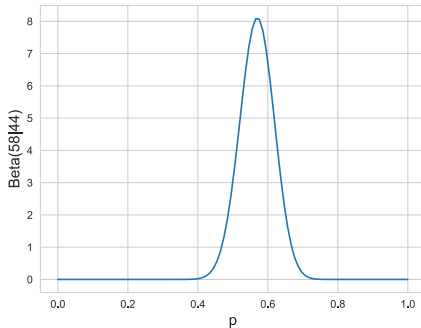
$$P(\Theta|x) = \frac{P(x|\Theta)}{P(x)} \cdot P(\Theta)$$

with $P(x) = \int_{\Theta} P(x|\Theta)P(\Theta)$ **marginal likelihood**

Notes:

Mean posterior estimation

- ▶ Bayesian inference yields **full posterior distribution** of model parameter p as
$$P(p|k) = \text{Beta}(p; x + 1, n - x + 1)$$
- ▶ this allows us to assign a **degree of belief** for any $p \in [0, 1]$
- ▶ but: we often want **point estimate** $\hat{p}$
- ▶ we can use **mean of posterior distribution** to estimate $\hat{p}$
- ▶ minimises **mean squared error** $E[(\Theta - \hat{\Theta})^2]$
- ▶ using mean $\frac{\alpha}{\alpha + \beta}$ of $\text{Beta}(\alpha, \beta)$ we find
$$\hat{p} = \frac{x + 1}{n + 2}$$



posterior distribution after observing $x = 57$ heads in $n = 100$ coin tosses

example: $n = 100$ tosses, $x = 57$ heads

$$P(p|x) = \text{Beta}(p; 58, 44), \hat{p} = \frac{58}{102} \approx 0.569$$

Notes:

Maximum a posteriori (MAP) estimation

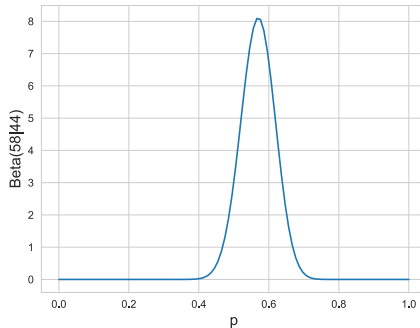
- ▶ instead of mean posterior, we can estimate $\hat{p}$ based on **maximum a posteriori probability** (MAP), i.e.

$$\begin{aligned}\hat{p} &= \operatorname{argmax}_p P(p|x) \\ &= \operatorname{argmax}_p \operatorname{Beta}(p; x + 1, n - x + 1)\end{aligned}$$

- ▶ using maximum $\frac{\alpha-1}{\alpha+\beta-2}$ of $\operatorname{Beta}(\alpha, \beta)$ we find

$$\hat{p} = \frac{x}{n}$$

- ▶ for uniform prior, MAP and MLE coincide
- ▶ but: Bayesian inference ...
 - ▶ yields full posterior distribution
 - ▶ allows to consider other prior distributions



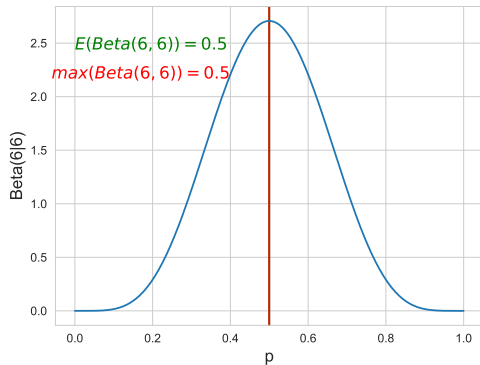
posterior distribution after observing $x = 57$
heads in $n = 100$ coin tosses

example: $n = 100$ tosses, $x = 57$ heads

$$P(p|x) = \operatorname{Beta}(p; 58, 44), \hat{p} = \frac{57}{100} = 0.57$$

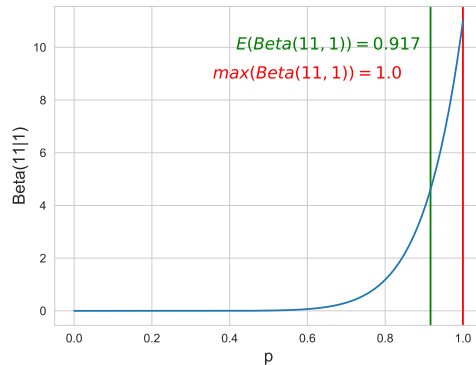
Notes:

Mean vs. maximum posterior estimates



$$n = 10, x = 5$$

$$P(p|x) = \text{Beta}(p; 6, 6)$$



$$n = 10, x = 10$$

$$P(p|x) = \text{Beta}(p; 11, 1)$$

When do mean and maximum a posterior estimators coincide?

Notes:

A Bayesian view on the sunrise problem

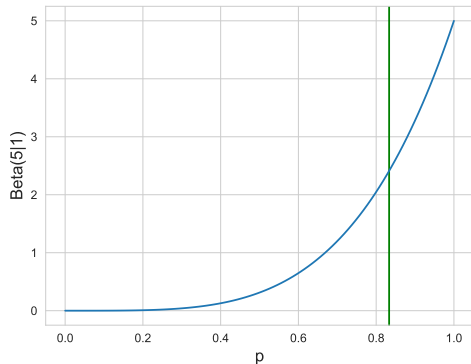
- ▶ we can reformulate the **sunrise problem** in terms of coin tosses, where heads corresponds to a sunrise
- ▶ we start with **uniform prior on p** , i.e. $\text{Beta}(p; 1, 1)$
- ▶ after observing $x = n$ heads in all n coin tosses the **posterior distribution** is given as

$$P(p|x) = \text{Beta}(p; n + 1, 1)$$

- ▶ **mean posterior estimate $\hat{p}$** yields

$$\hat{p} = \frac{\alpha}{\alpha + \beta} = \frac{n + 1}{n + 2}$$

- ▶ we recover **Laplace's rule of succession!**



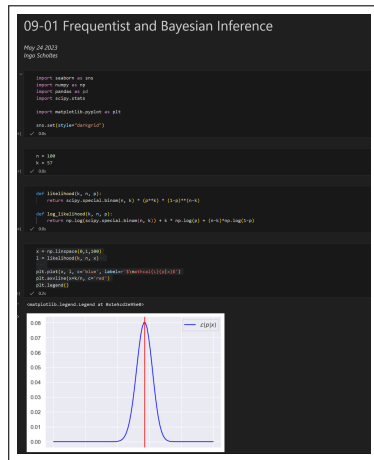
$$n = 5, \hat{p} = \frac{6}{7} \approx 0.85$$

posterior distribution after observation of five
sunrises

Notes:

Practice Session

- ▶ we implement **frequentist and Bayesian inference** in python
- ▶ we apply it to our **coin tossing example**
- ▶ we explore the **posterior distribution** for different numbers of observed coin tosses and outcomes



practice session

see notebook 11-01 in gitlab repository at

→ https://gitlab.informatik.uni-wuerzburg.de/ml4nets_notebooks/2025_sose_akids2

Notes:

Statistical inference and prediction

- for observations of multivariate RV (X, Y) , we are often interested in the **relationship between X and Y** , i.e. function f such that

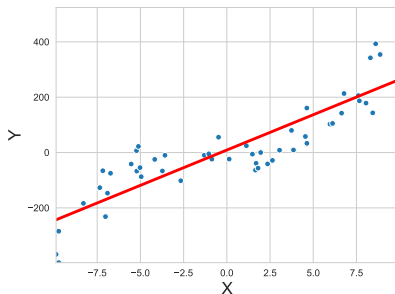
$$Y \approx f(X)$$

- for given f we typically have **error term ϵ** , i.e.

$$Y = f(X) + \epsilon$$

- example: least-squares linear regression
- we can use statistical model $P(Y|X)$ to **make predictions**, e.g.

$$f(X) := \underset{y}{\operatorname{argmax}} P(Y = y|X)$$



inference

- we want to **describe data with interpretable model**, e.g. by using linear model $f(X) = \beta_0 + \beta_1 X$
- we favor simple model (e.g. linear f) over small ϵ

prediction

- we want to **accurately predict** Y based on X
- we favor model that minimises error ϵ

Notes:

Linear regression

- consider **linear relationship** f between random variables X and Y , i.e.

$$Y = f(X) = \beta_0 + \beta_1 X$$

with **intercept** β_0 and **slope** β_1

- we call X **independent variable** and Y **dependent variable**

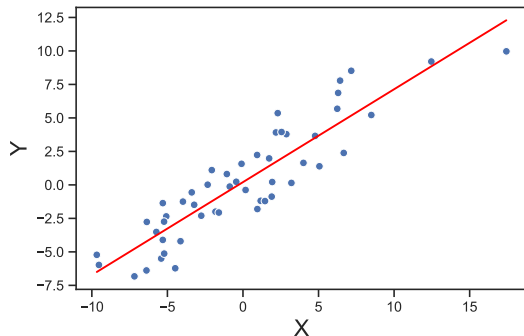
ordinary least squares regression

For realizations of bivariate random variable (X, Y) , **linear regression** is the process of finding coefficients β_0 and β_1 of a linear model

$$f(X) = \beta_0 + \beta_1 X$$

such that $\epsilon = Y - f(X)$ is minimised.

- how can we minimise error ϵ ?



Adrien-Marie Legendre

1752 – 1833



Carl Friedrich Gauss

1777 – 1855

Notes:

Residual sum of squares

- ▶ consider realisations $(x_i, y_i)_{i=1, \dots, n}$ of bivariate random variable (X, Y)
- ▶ for given coefficients β_0 and β_1 , linear model yields **prediction** $\hat{y}_i$ for each x_i as

$$\hat{y}_i := f(x_i) = \beta_0 + \beta_1 x_i$$

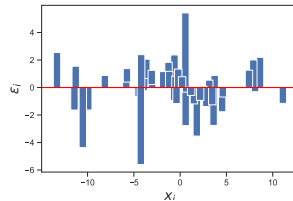
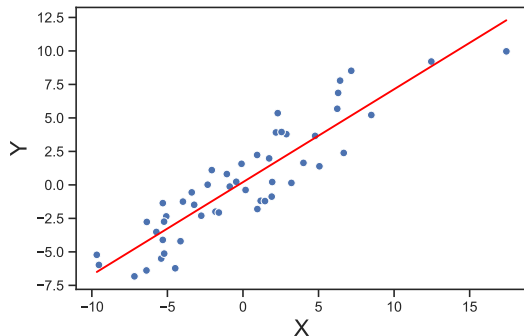
- ▶ i -th **residual** (“remainder”) is given as

$$\epsilon_i := y_i - \hat{y}_i$$

- ▶ we define **residual sum of squares** as

$$\text{RSS}(\beta_0, \beta_1) := \sum_{i=1}^n \epsilon_i^2$$

- ▶ RSS is **loss of model** with parameters β_0, β_1



$$\text{RSS}(-0.36, 0.76) = 178.44$$

Notes:

Parameter estimation

- coefficients that **best fit data** are

$$(\hat{\beta}_0, \hat{\beta}_1) = \underset{\beta_0, \beta_1}{\operatorname{argmin}} \operatorname{RSS}(\beta_0, \beta_1)$$

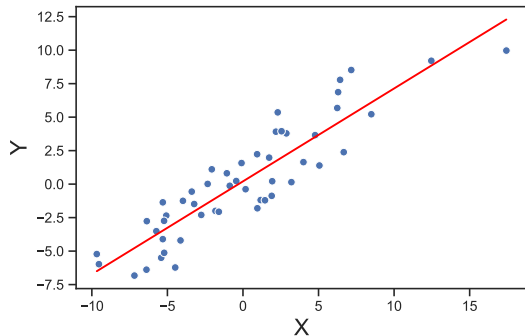
- we can **minimise RSS** using

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

with **sample means**

$$\bar{x} := \frac{1}{n} \sum_{i=1}^n x_i \quad \bar{y} := \frac{1}{n} \sum_{i=1}^n y_i$$



linear model resulting from ordinary **least squares** (OLS) regression

Notes:

Linear regression as statistical inference

- ▶ linear regression = **statistical model** for dependent variable Y given independent variable X
- ▶ for fixed X, β_0, β_1 let us assume $Y \sim \text{Norm}(\hat{Y}, \sigma)$ for $\hat{Y} = \beta_0 + \beta_1 X$, i.e. real value of Y is normally distributed around “predicted” value $\hat{Y}$
- ▶ equivalent to saying that $Y - \hat{Y} \sim \text{Norm}(0, \sigma)$, i.e. we assume **normally distributed residuals** with **homogeneous** σ
- ▶ assuming **i.i.d. residuals** we can write **likelihood function** of model parameters as

$$\mathcal{L}(\beta_0, \beta_1, \sigma | Y, X) = \prod_{i=1}^n \text{Norm}(\epsilon_i; 0, \sigma) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{\epsilon_i^2}{2\sigma^2}}$$

where $\epsilon_i := \beta_0 + \beta_1 x_i - y_i$ is the i -th residual

- ▶ we obtain the following **log-likelihood function**

$$\log \mathcal{L}(\beta_0, \beta_1, \sigma | Y, X) = -\frac{n}{2} \log \sigma^2 - \frac{n}{2} \log(2\pi) - \frac{1}{2\sigma^2} \sum_{i=1}^n \epsilon_i^2$$

Notes:

MLE perspective on linear regression

- ▶ consider log-likelihood function

$$\log \mathcal{L}(\beta_0, \beta_1, \sigma | Y, X) = -\frac{n}{2} \log \sigma^2 - \frac{n}{2} \log(2\pi) - \frac{1}{2\sigma^2} \sum_{i=1}^n \epsilon_i^2$$

- ▶ first two summands and factor $\frac{1}{2\sigma^2}$ can be treated as constants that do not depend on β_0 and β_1
- ▶ finding β_0, β_1 that maximise $\log \mathcal{L}(\beta_0, \beta_1, \sigma | Y, X)$ is equivalent to minimising

$$\sum_{i=1}^n \epsilon_i^2 = \text{RSS}(\beta_0, \beta_1)$$

- ▶ holds under the assumption $Y - \hat{Y} \sim \text{Norm}(0, \sigma)$, i.e. deviation of residuals $y_i - \hat{y}_i$ does not depend on x_i (we say that residuals are **homoskedastic**)

least squares regression can be interpreted as MLE of model parameters for **independent and homoskedastic residuals** that follow a **zero-centered normal distribution**

Notes:

Bayesian perspective on linear regression

- ▶ for given X, β_0, β_1 let $Y \sim \text{Norm}(\hat{Y}, \sigma)$ with $\hat{Y} := \beta_0 + \beta_1 \cdot X$
- ▶ for the **likelihood function** we found

$$\mathcal{L}(\beta_0, \beta_1, \sigma | Y, X) = P(Y, X | \beta_0, \beta_1, \sigma) = \prod_{i=1}^n \text{Norm}(\epsilon_i; 0, \sigma)$$

- ▶ for **posterior distribution** after seeing data (X, Y) we have

$$P(\beta_0, \beta_1, \sigma | Y, X) = \frac{P(Y, X | \beta_0, \beta_1, \sigma) \cdot P(\beta_0, \beta_1, \sigma)}{P(Y, X)}$$

- ▶ for **uniform priors** we have $P(\beta_0, \beta_1, \sigma) \equiv \text{const} \forall \beta_0, \beta_1, \sigma \in \mathbb{R}$ and thus

$$P(\beta_0, \beta_1, \sigma | Y, X) \propto P(Y, X | \beta_0, \beta_1, \sigma) = \mathcal{L}(\beta_0, \beta_1, \sigma | Y, X)$$

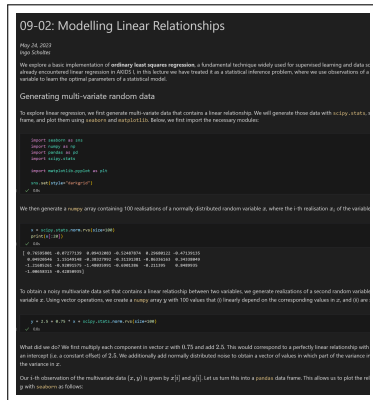
- ▶ for **uniform prior**, maximum a posterior (MAP) estimation is equivalent to MLE → slide 14

least squares regression can be interpreted as **Bayesian inference of parameters with uniform priors on β_0 and β_1** and independent and homoskedastic residuals that follow a zero-centered normal distribution

Notes:

Practice Session

- ▶ we implement **least squares linear regression** in python
- ▶ we calculate the **log-likelihood of model parameters**
- ▶ we show how to perform linear regression in the **machine learning library sklearn**
- ▶ we implement measures to assess the **accuracy of regression models**



practice session

see notebooks 11-02 and 11-03 in gitlab repository at

→ https://gitlab.informatik.uni-wuerzburg.de/ml4nets_notebooks/2025_sose_akids2

Notes:

In summary

- ▶ we can use frequentist and Bayesian techniques to **infer most plausible model parameters** given observed data
- ▶ Bayesian inference allows to incorporate **prior knowledge** about process that generates the data
- ▶ **posterior distribution** allows us to assess plausibility of any given model parameter
- ▶ statistical inference is foundation for **statistical learning, pattern recognition and probabilistic AI**
- ▶ **statistical hypothesis testing** enables us to formulate and assess **probabilistic propositions** → L12

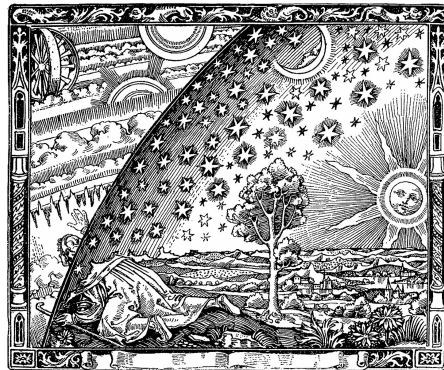


image credit: Camille Flammarion, L'Atmosphère: Météorologie Populaire, 1888, public domain

Notes:

Self-study questions

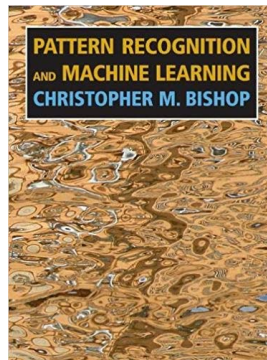
1. How do the frequentist and Bayesian interpretation of probabilities differ?
2. What is the role of a prior in the context of Bayesian inference?
3. What does Bayes theorem tell about the connection between prior and posterior probabilities.
4. Show how Laplace's rule of succession can be derived from Bayesian updating with a uniform prior.
5. How can we estimate model parameters using the posterior distribution?
6. Explain the difference between model likelihood and posterior probability in Bayesian inference.
7. Why can we use the mean of the posterior to estimate model parameters?
8. What is the difference between inference and prediction?
9. Define an ordinary linear regression model for a bivariate random variable (X, Y)
10. How are residuals defined and what is the residual sum of squares (RSS)?
11. Under which conditions does the minimisation of the RSS in linear regression corresponds to a maximum likelihood estimation of model parameters?
12. Under which conditions does the minimisation of the RSS in linear regression corresponds to a Bayesian inference of model parameters?
13. Under which condition does the MLE and MAP-based estimate of model parameters coincide?

Notes:

References

reading list

- ▶ Hasties: **Introduction to Statistical Learning**, Springer, 2006 → Chapters 1-3
- ▶ Kevin P. Murphy: **Machine Learning - A Probabilistic Perspective**
- ▶ Christopher M. Bishop: **Pattern Recognition and Machine Learning**, Springer, 2006, → Chapters 1-3
- ▶ Allen B. Downey: **Think Bayes**, 2012
- ▶ Clemens Ester et al.: **A Guide to Bayesian Inference for Regression Problems**, EMRP project NEW04 “Novel mathematical and statistical approaches to uncertainty evaluation”, 2015
- ▶ John K. Kruschke et al.: **The Time has Come: Bayesian Methods for Data Analysis in Organizational Sciences**, Organizational Research Methods, Vol. 15, 2012



Notes: